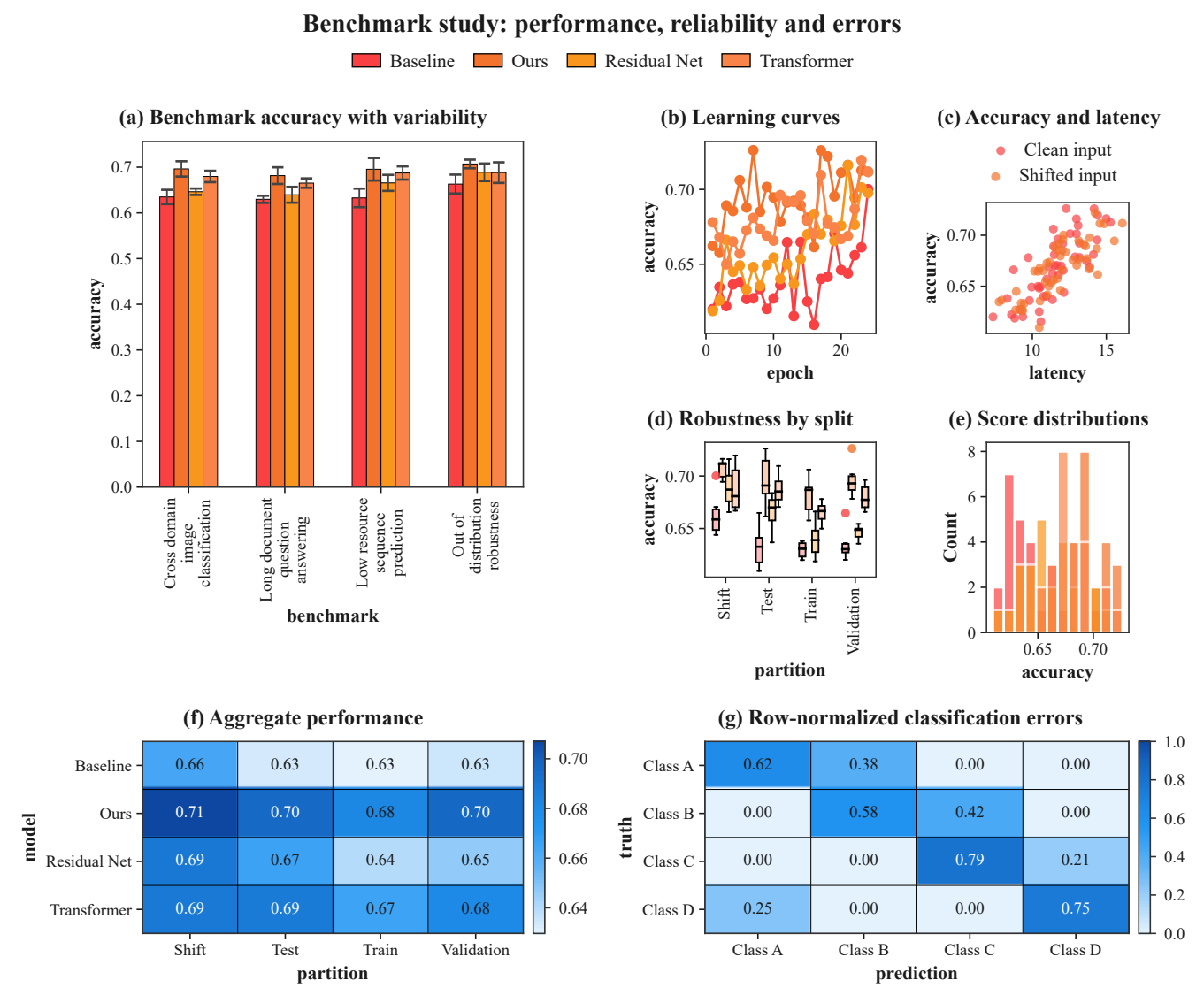

Benchmark study: performance, reliability and errors
Baseline
Ours
Residual Net
Transformer
(a) Benchmark accuracy with variability
accuracy
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Cross domain image classification
Long document question answering
Low resource sequence prediction
Out of distribution robustness
benchmark
(b) Learning curves
accuracy
0.65
0.70
0
10
20
epoch
(c) Accuracy and latency
Clean input
Shifted input
accuracy
0.65
0.70
10
15
latency
(d) Robustness by split
accuracy
0.65
0.70
Shift
Test
Train
Validation
partition
(e) Score distributions
Count
0
2
4
6
8
0.65
0.70
accuracy
(f) Aggregate performance
model
Baseline
Ours
Residual Net
Transformer
0.66
0.63
0.63
0.63
0.71
0.70
0.68
0.70
0.69
0.67
0.64
0.65
0.69
0.69
0.67
0.68
Shift
Test
Train
Validation
partition
0.64
0.66
0.68
0.70
(g) Row-normalized classification errors
truth
Class A
Class B
Class C
Class D
0.62
0.38
0.00
0.00
0.00
0.58
0.42
0.00
0.00
0.00
0.79
0.21
0.25
0.00
0.00
0.75
Class A
Class B
Class C
Class D
prediction
0.0
0.2
0.4
0.6
0.8
1.0